B.TECH CAPSTONE PROJECT • DEEP LEARNING • CONTINUOUS GLUCOSE MONITORING

HYPOGUARDIAN AI

Hypoglycemia Monitoring & 30-Minute Early Warning Using LSTM Networks

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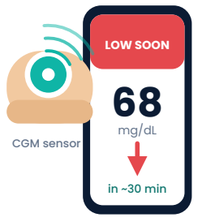
Project Guide: Dr. Umakant Mandawkar

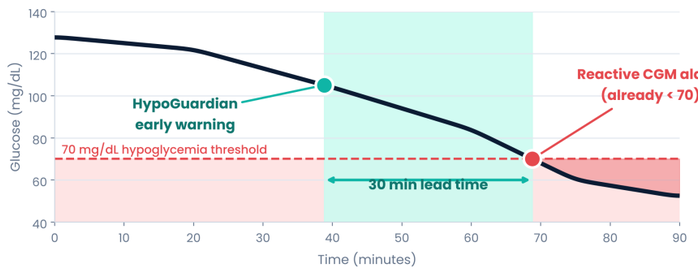
Symbiosis Institute of Technology, Nagpur • Dept. of CSE • AY 2025–26

01 THE CHALLENGE

Alarms react. Forecasting buys time.

- Hypoglycemia: glucose below **70 mg/dL** (3.9 mmol/L).
- Low glucose may cause shakiness, confusion and, in severe cases, seizures.
- CGM streams time-stamped readings, ideal for **time-series learning**.
- **Aim:** estimate if glucose will fall below 70 mg/dL within **30 minutes**.

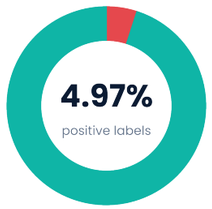




Conceptual illustration only — not patient data.

02 DATA & PREPROCESSING

6	85,225	4.97%
participants	CGM readings	positive labels



- **4,226 positive windows**
- **80,833 negative windows**

166 timestamps lacked future coverage. 2,736 readings (3.21%) were below 70 mg/dL.

- **Target:** positive if any reading in (t, t+30 min] is below 70 mg/dL.
- **Input:** 12 recent readings (about 1 h, mostly 5 min apart).
- Sequences stay inside participant and split boundaries; scaler is fitted on **training data only**.

03 MODEL ARCHITECTURE

An LSTM learns **how fast and in which direction** glucose is moving, not just its latest value.

INPUT	LSTM	DROPOUT	DENSE	DROPOUT	OUTPUT
12 × 1 glucose	64 units (gates)	0.30	32 ReLU	0.20	0–1 sigmoid

Output: probability of glucose below 70 mg/dL within 30 min

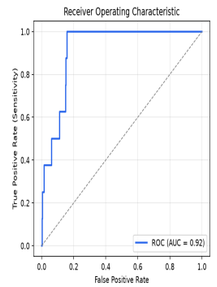
Adam • Binary cross-entropy • Class weighting • 19,009 params

PROJECT PIPELINE

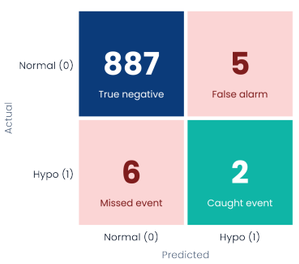
CGM XML → CLEAN → LABEL → WINDOW → LSTM → EVALUATE

04 RESULTS & HONEST EVALUATION

~0.980	~0.978	0.918
Training AUC	Random-window validation AUC	Chronological held-out ROC-AUC



ROC curve (AUC 0.92)



Confusion matrix (900 sequences)

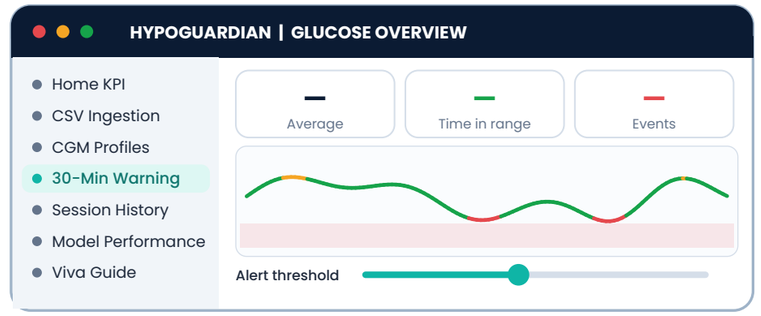
99.44%	25%	28.57%
specificity	sensitivity	precision

- **Random validation looked superb**, but adjacent windows share 11 of 12 readings, so ~0.978 is **optimistic**.
- **Held-out test:** only 2 of 8 events caught and PR-AUC is just 0.10, so sensitivity must improve.
- **98.78% accuracy misleads:** events are only ~0.9% of windows.

EVALUATION STATUS

Sensitivity is still low on a tiny event count. Participant-wise splits and more data come before any alerting.

05 DASHBOARD & LIVE APP



Illustrative interface graphic — not a live screenshot.

7 interactive pages • Threshold slider 0.1–0.9 • 16 pytest tests passed • Docker-ready

SCAN TO OPEN THE LIVE APP

Explore CGM profiles and the 30-minute early-warning demo on any phone, tablet or laptop.

hypoguardian-ai-m7nyvcy6n9scv2och7ri3p.streamlit.app

06 LIMITATIONS & NEXT STEPS

LIMITATIONS

- Six participants, one dataset: small cohort.
- Glucose-only input: no insulin, meals, activity or sleep.
- Class imbalance and overlapping-window leakage.
- No prospective clinical validation yet.

NEXT STEPS

- Leakage-resistant, participant-wise evaluation with baselines.
- Show model outputs only after suitable validation.
- Add data, more signals and expert review.

ACADEMIC RESEARCH PROTOTYPE ONLY

Not for diagnosis, treatment, emergency decisions or autonomous clinical alerts.